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MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAREXAMINATIONNOVEMBER2025

FIRSTSEMESTER

Multi-DisciplinaryCourses(MDC)-MG1MDCCFS100-FOUNDATIONOF CYBERFORENSICS

(2024ADMISSIONONWARDS)

Duration:1.5Hours **MaximumMarks:35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PartA

Short Answer Type Questions

Answer any **10** Questions

Each Question carries**1**mark

1 Volatile and non volatile data based on their permanence, or by their source and content, such as communication logs, user-created files, and system artifacts. Also deleted, hidden files, password-protected, and encrypted data. [An]/[CO2]

2 **Definition: the process of copying data from a primary location to a secondary, separate location to protect it from loss. (1 mark)** [An]/[CO2]

Backup Obstacles: Backup window, Network bandwidth, System throughput, Lack of resources

Factors affecting back-up: Storage costs are decreasing,
Systems have to be online continuously.
The role of backup has changed now (1 mark)

3 FTK Imager OR EnCase OR The Sleuth Kit (TSK) / Autopsy OR Forensic imager [U]/[]

4 Method 1: Create a desktop shortcut to the Registry Editor C:\\Windows\\regedit.exe [U]/[]

Method 2: Create a .reg file for a specific key (1 mark for each method)

5 Compare with their contributions in computer forensic technologies [An]/[CO2]

Military computer forensic technologies - eg.CFX-2000, (0.5 mark)
Law Enforcement computer forensic technologies -eg. Mirror Image Backup Software, Anadisk, Diskette Analysis Tool, Copyqm: Diskette Duplication Software ,Text Search Plus Intelligent Forensic Filter, (1 mark)
Business computer forensic technologies - eg. Virus/Trojan/Worm Protection(0.5 mark)

6. Problems with data recovery and factors need to concern for efficient back up [An]/[CO2]

Problems :Shrinking Expertise, Growing Complexity Failures, Budgets and Downtime (1 mark)

Solutions: prepare Before Back Up, Evaluate Preparations,-Automated Recovery, Make Recoveries Efficient, Take Backups (1 mark)

7. Raw (dd): extensions -dd, .img, .raw, .001 [U]/[]

EWF (Expert Witness Format): extensions- .E01, .Ex01

AFF (Advanced Forensic Format): extensions- .aff, .aff4

AccessData Custom Content Image: extensions -.AD1

SMART

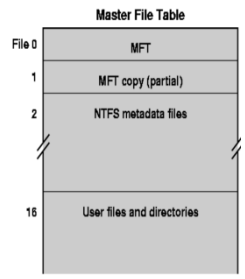
WinHex

8. To understand Registry Explorer, use its graphical interface to load registry "hives" from a file or live system to see a tree-like structure of keys and their values. You can navigate this structure to find information, view deleted keys and values, and analyze forensic data like user login times or recently used files [U]/[]

9 Unallocated space-free space on a disk, not been assigned to any file and can be used to store new data, (1 mark) [U]/[CO1, CO3]
Slack space - leftover, unused portion of a cluster that is already allocated to an existing file. (1 mark)

10 Backup Obstacles: Backup window, Network bandwidth, System throughput, Lack of resources (1 mark) [An]/[CO2]
Factors affecting back-up: Storage costs are decreasing,
Systems have to be online continuously.
The role of backup has changed nowadays (1 mark)

11. Contains metadata for every file and directory on a volume, acting as an index for all its contents. (1 mark) [U]/[]



(1 mark)

12. **For performance (1mark)** [U]/[CO4]

Registry cleanup- Use the Registry Editor to find and remove obsolete or redundant entries

that can cause performance issues.

Disable startup programs- disable programs that run automatically at startup.

Adjust system settings- Tune memory management or display settings, to optimize responsiveness.

For security (1 mark)

Restrict anonymous access**,** Change permissions

Disable features that could be exploited by malware or attackers

Prevent persistence: helps in remove malicious entries that allow malware to automatically run on startup.

Monitor for threats: help in detecting and removing persistent threats.

[10x1=10]

Part B

ShortEssayTypeQuestions

Answerany**3**Questions

EachQuestioncarries**5**marks

13. **Standard Service (2.5 mark)** [U]/[CO1, CO3]
Computer forensics experts should be able to work on your case during normal business hours.
On-Site Service (2.5 mark)
Computer forensics experts should be able to travel to the location. At the on-site, the experts should quickly be able to produce exact duplicates of the data storage media in question.

14. 1. Computer forensics is the process of methodically examining computer media (hard disks, diskettes, tapes, etc.) for evidence. (1mark) [An]/[CO2]

Computer forensics services (1.5 mark)

The computer forensic objective, priority, accuracy versus speed, the computer forensics specialist

2. Military computer forensic technology (1.mark)
Law enforcement computer forensic technology (1 mark)
Business computer forensic technology (0.5 mark)

15. FTK Imager allows for the preview, analysis, and export of files, including deleted ones, without altering the source data. (1 mark) [E]/[CO4]

Phase 1: Evidence Acquisition (Creating the Image) (2 mark)

Launch FTK Imager and Select Source: File > Create Disk Image

Configure Image Destination and Format: specify where the image file will be saved.
Enter Case Information: Fill in relevant details such as case number, examiner name etc.
Start Imaging and Verification

Phase 2: Evidence Examination and Extraction (2 mark)

Add Evidence Item: File > Add Evidence Item and select the created image file.

Navigate the File System: browse the file system hierarchy that includes visible files, hidden partitions, and unallocated space.

Identify Relevant Data from Deleted Files, Volatile Data, Encrypted Files

Extract Evidence: Right-click on the specific files or folders of interest and select

Export Files to save them to a different, secure location on the analysis workstation.

Document Findings: Maintain a detailed chain of custody and document all findings,

16. Intercept and log data packets moving across a network to provide visibility for analysis, troubleshooting, and security monitoring. (1 mark) [An]/[]

Working:- Packet interception, Data capture, Data decoding, Data presentation (1 mark)

Role:- Network Troubleshooting and Performance Optimization , Cyber security and Intrusion Detection, Incident Response and Forensic Analysis, Network Planning and Capacity Management, Compliance and Auditing. (2 mark)

Examples:- Wireshark, tcpdump, and Network Miner, Snort, Commercial Solutions (e.g., SolarWinds, NetWitness, Riverbed) (1 mark)

[3x5=15]

Part C

EssayTypeQuestions

Answerany**1**Question

EachQuestioncarries**10**marks

17. If the computer and its contents are examined by anyone other than a trained and experienced computer forensics specialist, the usefulness and credibility of that evidence will be tainted Look very carefully at the level of experience of the individuals involved while Choosing a Computer Forensics Specialist for a Criminal Case. (2.5 mark) [U]/[CO1, CO3]

COMPUTER FORENSICS SERVICES (2.5 mark)
Data seizure, Data duplication and preservation, Data recovery, Document searches, Media conversion, Expert witness services, Computer evidence service options
Other miscellaneous services
MEASURES (5 mark)
Collection and analysis : Forensically sound collection,Minimize alteration, Use validated tools,
Documentation and chain of custody Detailed logs, Chain of custody, Hashing or "fingerprint" of the data
Storage and access :Secure storage, Access control

Legal and expert support :Legal authority, Expert testimony

18 Specialized software and hardware used to identify, preserve, extract, analyze, and report on electronic evidence for legal purposes (definition 1 mark) [U]/[]

Data Acquisition and Preservation Tools :Hardware Write-Blockers (each categories 1 mark)

Forensic Imagers/Duplicators: FTK Imager dd or dc3dd
Data Analysis and Examination Tools: EnCase and Magnet AXIOM
File Analysis and Recovery Tools :The Sleuth Kit, Autopsy, Scalpel) ExifTool

Memory Analysis Tools: Volatility and MAGNET RAM

Network Analysis Tools: Wireshark

[1x10=10]